HW 1

Problem 1

1)

In-memory computing architectures such as analog and quantum computing can compute in-place, so it don't need to move data between CPU and memory like traditional von-neumann architecture. For example, analog in-memory computing do computation inside memory using continuous electrical signals.

2)

The heterogeneous architecture is most viable because no single hardware type is best for everything. The heterogeneous architecture can perform different tasks by proper components, so it can achieve better performance, scalability and energy efficiency.

Problem 2

1)

$$flop = 2n * n * n = 2n^3$$

2)

Assume each value has t bytes

$$minimum\ data\ move = (n^2 + n^2 + n^2) * t = 3n^2 * t$$

3)

$$I = \frac{2n^3}{3n^2 * t} = \frac{2n}{3t}$$

4)

For SpMV y ← Ax

$$flop = 2nnz(A)$$

Assume each index has s bytes, value has t bytes:

$$data = s * nnz(A) + t * nnz(A) = (s + t)nnz(A)$$

Then

$$I = \frac{2nnz(A)}{(s + t)nnz(A)} = \frac{2}{s + t}$$

5)

Assume s(index) = 4 bytes, t(bytes) = 8 bytes

For SpMV:

$$I = \frac{2}{12} = 0.167$$

I is tiny. Therefore; SpMV is memory-bound.

For GEMM:

$$I = \frac{2n}{3 * 8} = 0.167n$$

When n is big, the I can beyond 1, so GEMM can be compute-bound.

Problem 3

1)

Assume batch size is B, and the matrix size is n*n. Then:

$$flop = B * 2n * n * n = B * 2n^3$$

$$data = B * (n^2 + n^2 + n^2) * t = B * 3n^2 * t$$

$$I = \frac{B * 2n^3}{B * 3n^2 * t} = \frac{2n}{3t}$$

2)

Minimum intensity for Compute-bound is

$$i = \frac{10TFLOPS}{500GB} = 20$$

3)

For this gpu

$$ridge\ point = \frac{120TFLOPS}{3TB} = 40$$

It is harder to be “Compute bound”

4)

Assume s(index) = 4 bytes, t(bytes) = 8 bytes

For SpMV:

$$I = \frac{1}{12} = 0.167$$

It is memory-bound and memory bandwidth is 500GB/s, and the throughput is

$$throughput = 0.167 * 500GB/S = 83.5GFLOP/s$$

Problem 4

1)

1. During the stream of A, computer two sketches Y and W

$$Y = A\Omega, W = \Psi A$$

2. Q = orth(Y) using TSQR
3. Recover the small matrix without rereading A

$$W \approx (\Psi Q)B \Rightarrow B = (\Psi Q)\dagger W$$

4. Output rank-k approximation

2)

Form a small random test matrix

$$\Omega \in R^{n\times(k+p)}$$

Stream A once and compute the sketch

$$Y = A\Omega$$

Orthonormalize the sketch to get the subspace.

This produces Q in one pass , without needing $Q^T A$

Problem 5

1)

Because natural images have most of their energy in the top singular values.

Choosing a small rank k≪1000k keeps the dominant large-scale structures (edges, shapes) while dropping small singular values that mostly represent fine noise/details.

So A_k gives a much smaller, denoised approximation with little visible loss.

2)

If the first 50 singular values capture 90% of the energy, then the remaining error energy is 10%.

So the minimum rank is:

$$k = 50.$$

Problem 6

1)

To achieve accuracy ε:

$$\|e_k\| = \rho^k \|e0\| \leq \varepsilon$$

Then:

$$\rho^k \leq \frac{\varepsilon}{\|e_0\|}$$

$$klog\rho \leq log(\frac{\varepsilon}{||e0||})$$

Since 0<ρ<1, we have logρ<0, so

$$k \geq \frac{log(\varepsilon/||e0||)}{logp}$$

2)

Wait-for-all:

$$K_A = \frac{log(\varepsilon/||e0||)}{logp}$$

$$Time_A = K_A T_s$$

Skip-stragglers:

$$K_B = \frac{log(\varepsilon/||e0||)}{logp_{eff}}, p_{eff} = 1 - q(1-p)$$

$$Time_B = K_B \tau$$

3)

if Time A > Time B, then:

$$\frac{log(\varepsilon/||e0||)}{logp_{eff}}\tau < \frac{log(\varepsilon/||e0||)}{logp}T_s$$

So:

$$\frac{\tau}{logp_{eff}} < \frac{T_s}{logp}$$

4)

(a)

$$p_{eff} = 1 - q(1-p) = 1 - 0.8(1-0.9) = 0.92$$

(b)

–

$$\frac{\tau}{logp_{eff}} = 1/|log0.92| = 11.99$$

$$\frac{T_s}{logp} = 5/log|0.9| = 47.46$$

(c)

skipping is faster

5)

When waiting need far more time than skipping in each iteration , and the fraction of workers that finish on time is high, skipping a good idea. Otherwise, waiting is better.